

MATH 113: Introduction to Abstract Algebra

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Chapter 1: Introduction to Groups

1.1 Basic Axioms and Examples

Definition: (1) A *binary operation* $*$ on a set G is a function $*$: $G \times G \rightarrow G$. For any $a, b \in G$ we shall write $a * b$ for $*(a, b)$. (2) A binary operation $*$ on a set G is *associative* if for all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$. (3) If $*$ is a binary operation on a set G we say elements a and b of G *commute* if $a * b = b * a$. We say $*$ (or G) is *commutative* if for all $a, b \in G$, $a * b = b * a$.

Definition: (1) A *group* is an ordered pair $(G, *)$ where G is a set and $*$ is a binary operation on G satisfying the following axioms: (i) $*$ is associative, (ii) $\exists e \in G$, denoted the *identity* of G , such that $\forall a \in G$, $a * e = e * a = a$, and (iii) $\forall a \in G$, $\exists a^{-1} \in G$ called the *inverse* of a such that $a * a^{-1} = a^{-1} * a = e$.

Proposition 1: If G is a group under the operation $*$, then (1) the identity of G is unique, (2) $\forall a \in G$, a^{-1} is uniquely determined, (3) $(a^{-1})^{-1} = a$, $\forall a \in G$, (4) $(a * b)^{-1} = (b^{-1}) * (a^{-1})$, and (5) for any $a_1, \dots, a_n \in G$, the value of $a_1 * a_2 * \dots * a_n$ is independent of how the expression is bracketed (this is called the *generalized associative law*).

Proposition 2: Let G be a group and let $a, b \in G$. The equations $ax = b$ and $ya = b$ have unique solutions for $x, y \in G$. In particular, the left and right cancellation laws hold in G , i.e. (1) if $au = av$, then $u = v$, and (2) if $ub = vb$, then $u = v$.

Definition: For G a group and $x \in G$, define the *order* of x to be the smallest positive integer n such that $x^n = 1$, and denote this integer by $|x|$. In this case x is said to be of order n . If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order.

Definition: Let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group with $g_1 = 1$. The *multiplication table* or *group table* is the $n \times n$ matrix whose i, j entry is the group element that results from the product $g_i g_j$.

1.2 Dihedral Groups

N/A

1.3 Symmetric Groups

N/A

1.4 Matrix Groups

Definition: (1) A *field* is a set F together with two commutative binary operations $+$ and $\cdot$ on F such that $(F, +)$ is an abelian group (call its identity 0) and $(F - \{0\}, \cdot)$ is also an abelian group, and the following *distributive law* holds: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$, $\forall a, b, c \in F$. (2) For any field F let $F^\times = F - \{0\}$.

1.5 Quaternion Group

N/A

1.6 Homomorphisms and Isomorphisms

Definition: Let $(G, *)$ and $(H, \circ)$ be groups. A map $\varphi : G \rightarrow H$ such that $\varphi(x * y) = \varphi(x) \circ \varphi(y)$, $\forall x, y \in G$, is called a *homomorphism*.

Definition: The map $\varphi : G \rightarrow H$ is called an *isomorphism* and G and H are said to be *isomorphic* or of the same *isomorphism type*, written $G \cong H$, if (1) φ is a homomorphism (i.e. $\varphi(xy) = \varphi(x)\varphi(y)$), and (2) φ is a bijection.

Fact: φ is an isomorphism $\iff$ there exists a well defined inverse homomorphism φ^{-1} .

Fact: If φ is an isomorphism, then: $|G| = |H|$, G is abelian $\iff H$ is abelian, $\forall x \in G$ it is true $|x| = |\varphi(x)|$.

1.6, 1.7 Homomorphisms, Group Actions

Definition: A *group action* of a group G on a set A is a map from $G \times A \rightarrow A$ satisfying (1) $g_1(g_2a) = (g_1g_2)a$, $\forall g_1, g_2 \in G$ and $\forall a \in A$, and (2) $1 \cdot a = a$, $\forall a \in A$.

Fact: For all fixed $g \in G$, $\exists \sigma_g$, a permutation of A where the map $g \mapsto \sigma_g$ is a homomorphism.

Chapter 2: Subgroups

2.1 Subgroups

Definition: Let G be a group. $H \leq G$ if H is nonempty and H is closed under products and inverses.

Proposition 1: (Subgroup Criterion) $H \leq G$ if $H \neq \emptyset$ and $xy^{-1} \in H, \forall x, y \in H$. If H is finite, it suffices to check that $H \neq \emptyset$ and H is closed under multiplication.

2.2 Centralizers, Normalizers, Stabilizers, Kernels

Definition: (Centralizer) $C_G(A) = \{g \in G \mid gag^{-1} = a, \forall a \in A\}$. It is the set of elements of G which commute with all elements of A .

Definition: (Center) $Z(G) = \{g \in G \mid gxg^{-1} = x, \forall x \in G\}$. It is the set of elements of G which commute with all elements of G .

Definition: (Normalizer) $N_G(A) = \{g \in G \mid gAg^{-1} = A\}$. It is the set of all elements of G which conjugate A .

Definition: (Stabilizer) If G acts on S and $s \in S$ is fixed, $G_s = \{g \in G \mid g \cdot s = s\}$.

Definition: (Kernel) If G acts on S , $\ker = \{g \in G \mid g \cdot s = s, \forall s \in S\}$.

* Note that all of these are subgroups of G .

2.3 Cyclic Groups and Cyclic Subgroups

Definition: A group H is cyclic if $\exists x \in H$ such that $H = \{x^n \mid x \in \mathbb{Z}\} = \langle x \rangle$.

Proposition 2: If $H = \langle x \rangle$, then $|H| = |x|$. Also, $|H| = \infty \implies x^a \neq x^b$.

Proposition 3: Let G be a group, $x \in G$, and $m, n \in \mathbb{Z}$. Then $x^n = x^m = 1 \implies x^{(m,n)} = 1$. Also, $x^m = 1 \implies |x| \mid m$.

Theorem 4: Two cyclic groups of the same order are isomorphic. In particular, if $|\langle x \rangle| = n < \infty$, $\langle x \rangle \cong \mathbb{Z}^n$, and if $|\langle x \rangle| = \infty$, $\langle x \rangle \cong \mathbb{Z}$.

Proposition 5: Let G be a group, $x \in G$, and $a \in \mathbb{Z} - \{0\}$. $|x| = \infty \implies |x^n| = \infty$ $|x| = n < \infty \implies |x^a| = \frac{n}{(n,a)}$.

Proposition 6: Let $H = \langle x \rangle$. $|x| = \infty \implies H = \langle x^a \rangle \iff a = \pm 1$ $|x| = n < \infty \implies H = \langle x^a \rangle \iff (a, n) = 1$.

Theorem 7: Let $H = \langle x \rangle$. Then: Every subgroup of H is cyclic. $|H| = \infty \implies \langle x^a \rangle \neq \langle x^b \rangle, \forall a, b$ such that $a \neq b$. $|H| = n < \infty \implies \forall a$ such that $a \mid n$ there is a unique subgroup of H of order a that is $\langle x^{\frac{n}{a}} \rangle$. Also, $\forall m \in \mathbb{Z}, \langle x^m \rangle = \langle x^{(n,m)} \rangle$.

2.4 Subgroups Generated by Subsets of a Group

Proposition 8: Let $\mathcal{A}$ be a nonempty collection of subgroups of G . Let $K = \bigcap_{H \in \mathcal{A}} H$. Then $K \leq G$.

Proposition 9: $\overline{A} = \{a_1^{\epsilon_1} \dots a_n^{\epsilon_n} \mid n \in \mathbb{Z}, n \geq 0, a_i \in A, \epsilon_i = \pm 1\} = \langle A \rangle$.

Chapter 3: Quotient Groups and Homomorphisms

3.1 Quotient Groups

Proposition 1: Let G, H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Then,

$$\varphi(1_G) = 1_H \quad \varphi(g^{-1}) = \varphi(g)^{-1} \quad \varphi(g^n) = \varphi(g)^n \quad \ker \varphi \leq G \quad \text{im}(\varphi) \leq H$$

Theorem 3: Let G, H be groups and $K = \ker \varphi$ with $\varphi : G \rightarrow H$. Then the set of left cosets of K in G is G/K and the operation $uK \circ vK = (uv)K$ is well defined if the converse of injectivity is true, i.e. $u = v \implies \varphi(u) = \varphi(v)$.

Proposition 4: Let G be a group and $N \leq G$. Then the set of left cosets of N in G partition G , and $\forall u, v \in G, uN = vN \iff v^{-1}u \in N$.

Proposition 5: Let G be a group and $N \leq G$. The operation $uN \cdot vN = (uv)N$ is well defined $\iff gNg^{-1} \in N, \forall g \in G$ and $\forall n \in N$. If the operation is well defined, the set of left cosets of N in G is a group.

Definition: gNg^{-1} is the conjugate of $n \in N$ by $g \in G$. The set $gNg^{-1} = \{gng^{-1} \mid n \in N\}$ is called the *conjugate* of N by g . The element g *normalizes* N if $gNg^{-1} = N$. A subgroup N of G is normal ($N \trianglelefteq G$) if $gNg^{-1} = N, \forall g \in G$. All subgroups of an *abelian* group are normal.

Theorem 6: Let $N \leq G$. The following statements are equivalent:

$$N \trianglelefteq G \quad N_G(N) = G \quad gN = Ng, \forall g \in G \quad \text{The set of left cosets of } N \text{ in } G \text{ are a group} \quad gNg^{-1} \subseteq N, \forall g \in G$$

Proposition 7: $N \trianglelefteq G \iff \exists \varphi$ such that φ is a homomorphism and $\ker \varphi = N$.

Definition: Let $N \trianglelefteq G$. The homomorphism $\pi : G \rightarrow G/N$ defined by $\pi(g) = gN$ is a natural projection/homomorphism. If $\bar{H} \leq G/N$, the complete preimage of $\bar{H}$ in G is the preimage of $\bar{H}$ under the natural projection

Fact: The quotient groups of a cyclic group are cyclic with order $\frac{|G|}{|N|}$.

3.2 Lagrange's Theorem

Theorem 8: (Lagrange) If G is a finite group and $H \leq G$, then $|H| \mid |G|$ and $|G : H| = \frac{|G|}{|H|}$. This also implies $\frac{|G:H|}{|H:K|} = |G : K|$.

Corollary 9: If G is a finite group and $x \in G$, $|x| = |\langle x \rangle| \mid |G|$ and $x^{|G|} = 1$.

Corollary 10: If G is a group of prime order p , then G is cyclic $\implies G \cong \mathbb{Z}_p$.

Theorem 11: (*Cauchy*) If G is a finite group and $p \mid |G|$, then $\exists x \in G$ such that $|x| = p$.

Theorem 12: (*Sylow*) If G is a finite group of order $p^\alpha m$, where p is a prime and $p \nmid m$, then $\exists H \leq G$ such that $|H| = p^\alpha$.

Definition: Let $H, K \leq G$. Then $HK = \{hk \mid h \in H, k \in K\}$.

Proposition 13: If $H, K \leq G$ and are finite, then $|HK| = \frac{|H||K|}{|H \cap K|}$.

Proposition 14: If $H, K \leq G, HK \leq G \iff HK = KH$.

Corollary 15: If $H, K \leq G$ and $H \leq N_G(K)$, then $HK \leq G$. In particular, if $K \trianglelefteq G$ then $HK \leq G$ for any $H \leq G$.

3.3 Isomorphism Theorems

Theorem 16: (1) If $\varphi : G \rightarrow H$ is a homomorphism of groups, then $\ker \varphi \trianglelefteq G$ and $G/\ker \varphi \cong \varphi(G)$.

Corollary 17: Let $\varphi : G \rightarrow H$ be a homomorphism of groups. φ is injective $\iff \ker \varphi = 1$, and $|G : \ker \varphi| = |\varphi(G)|$.

Theorem 18: (2) Let G be a group, let $A, B \leq G$ and assume $A \leq N_G(B)$. Then AB is a subgroup of G , $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$ and $AB/B \cong A/A \cap B$.

Theorem 19: (3) Let G be a group and let $H, K \trianglelefteq G$ with $H \leq K$. Then $K/H \trianglelefteq G/H$ and $(G/H)/(K/H) \cong G/K$.

Theorem 20: (4) Let G be a group and let $N \trianglelefteq G$. Then there is a bijection from the set of subgroups A of G which contain N onto the set of subgroups $\bar{A} = A/N$ of G/N . In particular, every subgroup of $\bar{G}$ is of the form A/N for some subgroup A of G containing N (namely, its preimage in G under the natural projection homomorphism from G to G/N). $\forall A, B \leq G$ with $N \leq A$ and $N \leq B$, this bijection has the following properties:

$$A \leq B \iff \bar{A} \leq \bar{B} \quad A \leq B \implies |B : A| = |\bar{B} : \bar{A}| \quad \overline{\langle A, B \rangle} = \langle \bar{A}, \bar{B} \rangle \quad \overline{A \cap B} = \bar{A} \cap \bar{B} \quad A \trianglelefteq G \iff \bar{A} \trianglelefteq \bar{G}$$

3.4 Composition Series and the Hölder Program

Proposition 21: If G is a finite abelian group and p is a prime dividing $|G|$, then $\exists x \in G$ with $|x| = p$.

Definition: A (finite or infinite) group G is *simple* if $|G| > 1$ and the only normal subgroups of G are 1 and G .

Definition: In a group G a sequence of subgroups $1 = N_0 \leq N_1 \leq \dots \leq N_{k-1} \leq N_k = G$ is called a *composition series* if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group for $0 \leq i \leq k-1$. If the above sequence is a composition series, the quotient groups N_{i+1}/N_i are called *composition factors* of G .

Theorem 22: (Jordan-Hölder) Let G be a finite group with $G \neq 1$. Then (1) G has a composition series and (2) The composition factors in a composition series are unique, namely, if $1 = N_0 \leq N_1 \leq \dots \leq N_r = G$ and $1 = M_0 \leq M_1 \leq \dots \leq M_s = G$ are two composition series for G , then $r = s$ and there is some permutation π of $\{1, 2, \dots, r\}$ such that $M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1}$, for $1 \leq i \leq r$.

Theorem: There is a list consisting of 18 (infinite) families of simple groups and 26 simple groups not belonging to these families (the *sporadic*

simple groups) such that every finite simple group is isomorphic to one of the groups in this list.

Theorem: (*Feit-Thompson*) If G is a simple group of odd order, then $G \cong Z_p$ for some prime p .

Definition: A group G is *solvable* if there exists a chain of subgroups $1 = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_s = G$ such that G_{i+1}/G_i is abelian for $i = 0, 1, \dots, s-1$.

Theorem: The finite group G is solvable $\iff$ for every divisor n of $|G|$ such that $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n .

3.5 Transpositions and the Alternating Group

Definition: A 2-cycle is called a *transposition*.

Definition: (1) $\epsilon(\sigma)$ is called the *sign of σ* . (2) σ is called an *even permutation* if $\epsilon(\sigma) = 1$ and an *odd permutation* if $\epsilon(\sigma) = -1$.

Proposition 23: The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism (where $\{\pm 1\} \cong Z_2$)

Proposition 24: Transpositions are all odd permutations and ϵ is surjective.

Definition: The *alternating group of degree n* , denoted by A_n , is the kernel of the homomorphism ϵ (i.e. the set of even permutations).

Proposition 25: The permutation σ is odd $\iff$ the number of cycles of even length in its cycle decomposition is odd.

Chapter 4: Group Actions

4.1 Group Actions and Permutation Representations

Definition: The *kernel* and *stabilizer* definitions follow similarly from 2.2. An action is *faithful* if its kernel is the identity.

Definition: The kernel of an action is the kernel of its permutation representation. In particular, an action of G on A induces a faithful action of $G/\ker \varphi$ on A , since $g, h \in G$ induce the same permutation on $A \iff g, h$ are representatives of the same coset of $\ker \varphi$. Note that $\forall a \in A$, the kernel of the action is contained in G_a , i.e. $\ker \varphi = \bigcap_{a \in A} G_a$. Conversely, given a set $A \neq \emptyset$ and any homomorphism φ of the group G into S_A we obtain an action of G on A such that $g \cdot a = \varphi(g)(a)$. The kernel of this action is the same as $\ker \varphi$. The permutation representation associated to this action is precisely the given homomorphism φ .

Proposition 1: For any group G and any nonempty set A there is a bijection between the actions of G on A and the homomorphisms of G into S_A .

Definition: If G is a group, a *permutation representation* of G is any homomorphism of G into the symmetric group S_A for some nonempty set A . We shall say a given action of G on A affords or induces the associated permutation representation of G .

Proposition 2: Let G be a group acting on $A \neq \emptyset$. The relation on A defined by $a \sim b \iff a = g \cdot b$ for some $g \in G$ is an equivalence relation. $\forall a \in A$, the number of elements in the equivalence class containing a is $|G : G_a|$.

Definition: Let G be a group acting on $A \neq \emptyset$. The equivalence class $\{g \cdot a \mid g \in G\}$ is the *orbit* of G containing a . The action of G on A is called *transitive* if there is only one orbit for the whole set A , i.e., $\forall a, b \in A, \exists g \in G$ such that $a = g \cdot b$.

Definition: Subgroups of symmetric groups are called *permutation groups*.

4.2 Groups Acting on Themselves by Left Multiplication (*Cayley's Theorem*)

Theorem 3: Let G be a group, $H \leq G$, and let G act by left multiplication on the set A of left cosets of H in G . Let π_H be the associated permutation representation afforded by this action. Then, G acts transitively on A ; G_{1H} is the subgroup H ; the action's kernel (i.e. $\ker \pi_H$) is $\bigcap_{x \in G} xHx^{-1}$ and $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Corollary 4: (*Cayley's*) A group G of order n is isomorphic to a subgroup of S_n .

Corollary 5: If G is a finite group of order n and p is the smallest prime such that $p \mid n$, then any subgroup of index p is normal. (Note, however, that a group of order n does not necessarily have a subgroup of index p).

4.3 Groups Acting on Themselves by Conjugation (*Class Equation*)

Definition: Two elements $a, b \in G$ are *conjugate* if $\exists g \in G$ such that $b = gag^{-1}$, i.e. they are in the same orbit of G acting on itself by conjugation. These orbits are called *conjugacy classes*.

Definition: Two subsets $S, T \subseteq G$ are *conjugate* if $\exists g \in G$ such that $T = gSg^{-1}$.

Proposition 6: The number of conjugates of a subset $S \subseteq G$ (order of its orbit) is $|G : N_G(S)|$. The number of conjugates of an element $s \in G$ is $|G : C_G(s)|$.

Theorem 7: (*Class Equation*) Let G be a finite group and $g_1, g_2, \dots, g_r$ be *representatives* of the distinct conjugacy classes of G that are not in the center $Z(G)$. Then $|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$.

Theorem 8: If p is a prime and P is a group of order p^α then $Z(P) \neq 1$, i.e. is nontrivial.

Corollary 9: If $|P| = p^2$ for some prime p , then P is abelian, and either $P \cong Z_{p^2}$ or $P \cong Z_p \times Z_p$.

Proposition 10: Let $\sigma, \tau \in S_n$ and let σ have cycle decomposition $(a_1 \dots a_{k_1})(b_1 \dots b_{k_2}) \dots$. Then $\tau\sigma\tau^{-1}$ has cycle decomposition $(\tau(a_1) \dots \tau(a_{k_1}))(\tau(b_1) \dots \tau(b_{k_2})) \dots$.

Definition: If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1, \dots, n_r$ with $n_1 \leq \dots \leq n_r$, then $n_1, \dots, n_r$ is the *cycle type* of σ . If $n \in \mathbb{Z}^+$. A partition of n is any nondecreasing sequence of positive integers whose sum is n .

Proposition 11: Two elements of S_n are conjugate in $S_n \iff$ they have the same cycle type.

Fact: If $H \trianglelefteq G$, then $\forall \mathcal{K}$ conjugacy classes of G , either $\mathcal{K} \subseteq H$ or $\mathcal{K} \cap H = \emptyset$.

Theorem 12: A_5 is a simple group.

Fact: Conjugation is often written as a right group action with notation $a^g = g^{-1}ag$.

4.4 Automorphisms

Definition: Let G be a group. An isomorphism from G onto itself is called an *automorphism* of G , the group of which is denoted $\text{Aut}(G)$. $\text{Aut}(G) \leq S_G$.

Proposition 13: Let $H \trianglelefteq G$. Then G acts by conjugation on H as automorphisms of H , i.e. the action of G on H by conjugation is an automorphism $\forall g \in G$. The permutation representation afforded by this action is a homomorphism of G into $\text{Aut}(H)$ with kernel $C_G(H)$. In particular, $G/C_G(H) \cong F$ such that $F \leq \text{Aut}(H)$.

Fact: Automorphisms must send subgroups to subgroups of the same order, elements of order n to elements of order n , etc.

Corollary 14: If $K \leq G$ and $g \in G$, then $K \cong gKg^{-1}$. Conjugate elements and conjugate subgroups have the same order.

Corollary 15: $\forall H \leq G, N_G(H)/C_G(H) \cong F$ such that $F \leq \text{Aut}(H)$. In particular, $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$ (consider

$H = G$).

Definition: Let G be a group and $g \in G$. Conjugation by g is called an **inner automorphism** of G and the subgroup of $\text{Aut}(G)$ containing all inner automorphisms is denoted by $\text{Inn}(G)$.

Definition: Let $H \leq G$. Then H is called **characteristic** in G , denoted $H \text{ char } G$, if $\forall \sigma \in \text{Aut}(G)$ we have $\sigma(H) = H$.

Fact: (1) Characteristic subgroups are normal. (2) If H is the unique subgroup of G for a given order, then $H \text{ char } G$. (3) If $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$, i.e. normality is transitive in this case.

Proposition 16: $\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group of order $\varphi(n)$.

Fact: Let G be a group with $|G| = pq$, with p, q prime, $p < q$, and $p \nmid q - 1$. Then, G is abelian.

Proposition 17: (1) If p is an odd prime and $n \in \mathbb{Z}^+$, $\text{Aut}(Z_{p^n})$ is cyclic with order $\varphi(p^n) = p^{n-1}(p - 1)$. (2) $\forall n \geq 3$, $\text{Aut}(Z_{2^n}) \cong Z_2 \times Z_{2^{n-2}}$, and is not cyclic but has a cyclic subgroup of index 2. (3) Let p be prime and let V be an abelian group (written additively) with $pv = 0, \forall v \in V$. If

$|V| = p^n$, then V is an n -dimensional vector space over the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, and $\text{Aut}(V) \cong GL(V) \cong GL_n(\mathbb{F}_p)$. (4) $\forall n \neq 6$,

$\text{Aut}(S_n) = \text{Inn}(S_n) \cong S_n$. For $n = 6$, $|\text{Aut}(S_6) : \text{Inn}(S_6)| = 2$. (5) $\text{Aut}(D_8) \cong D_8$ and $\text{Aut}(Q_8) \cong S_4$.

Fact: The V described in Prop. 17 is called the **elementary abelian group** of order p^n . For any prime p , $V_{p^2} = Z_p \times Z_p$, and $\text{Aut}(V_{p^2}) \cong GL_2(\mathbb{F}_p)$ has order $p(p - 1)^2(p + 1)$. Cor. 9 thus implies that $|P| = p^2 \implies |\text{Aut}(P)| = p(p - 1)$ or $p(p - 1)^2(p + 1)$, dependent on if P is cyclic or elementary abelian, respectively.

4.5 Sylow's Theorem

Theorem 18: (Sylow's) Let G be a group of order $p^\alpha m$, where p is a prime with $p \nmid m$. Then, (1) $\text{Syl}_p(G) \neq \emptyset$. (2) If $P \in \text{Syl}_p(G)$ and Q is any p -subgroup of G , $\exists g \in G$ such that $Q \leq gPg^{-1}$. In particular, any two Sylow p -subgroups of G are conjugate in G . (3) $n_p = |\text{Syl}_p(G)| \equiv 1 \pmod{p}$.

Moreover, $n_p = |G : N_G(P)|$ for any $P \in \text{Syl}_p(G)$, and thus $n_p \mid m$.

Lemma 19: Let $P \in \text{Syl}_p(G)$. If Q is any p -subgroup of G , then $Q \cap N_G(P) = Q \cap P$.

Corollary 20: Let P be a Sylow p -subgroup of G . Then the following are equivalent: (1) $\text{Syl}_p(G) = \{P\}$, i.e. $n_p = 1$. (2) $P \trianglelefteq G$. (3) $P \text{ char } G$. (4)

All subgroups generated by elements of p -power order are p -groups, i.e., if $X \subseteq G$ with $|x|$ being a power of p for all $x \in X$, then $\langle X \rangle$ is a p -group.

Facts: (1) $|G| = pq$ and $p \nmid q - 1 \implies$ abelian. If $P \in \text{Syl}_p(G)$ is normal in G , then G is cyclic. (2) $|G| = 30 \implies \exists H \trianglelefteq G$ with $H \cong Z_{15}$. (3)

$|G| = 12 \implies n_3 = 1$ or $G \cong A_4$. (4) $|G| = p^2q \implies \exists H \trianglelefteq G$ with $|H| = p$ or q .

Proposition 21: If $|G| = 60$ and $|\text{Syl}_5(G)| > 1$, then G is simple.

Corollary 22: A_5 is simple.

Proposition 23: If G is a simple group of order 60, then $G \cong A_5$.

(4.6) **Theorem 24:** A_5 is simple $\forall n \geq 5$.

Chapter 5: Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

Proposition 1: If $G_1, \dots, G_n$ are groups, $|G_1 \times \dots \times G_n| = |G_1| \dots |G_n|$. Note that if $|G_i| = \infty$, for any i , the direct product's order is similarly infinite.

Proposition 2: Let $G = G_1 \times \dots \times G_n$. (1) For each fixed i , the set of elements which have the identity of G_j in the j th position for all $j \neq i$ is a subgroup of G isomorphic to $G_i \cong \{(1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$. (2) For each fixed i , define $\pi_i : G \rightarrow G_i$ by $\pi_i((g_1, \dots, g_n)) = g_i$. Then π_i is a surjective homomorphism with $\ker \pi_i \cong G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n$.

Definition: The *elementary abelian group* is $E_{p^n} = \underbrace{Z_p \times \dots \times Z_p}_n$, with the property that $x^p = 1, \forall x \in E_{p^n}$.

5.2 The Fundamental Theorem of Finitely Generated Abelian Groups

Definition: (1) A group G is *finitely generated* if $\exists A \subseteq G$ with $|A|$ finite such that $G = \langle A \rangle$. (e.g. $\mathbb{Z}$ is finitely generated because $\mathbb{Z} = \langle 1 \rangle$). (2) For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \underbrace{\mathbb{Z} \times \dots \times \mathbb{Z}}_r$, where $\mathbb{Z}^0 = \mathbf{1}$. The group $\mathbb{Z}^r$ is called the *free abelian group of rank r* .

Theorem 3: (*Chapter Title*) Let G be a finitely generated abelian group. (1) Then $G \cong \mathbb{Z}^r \times Z_{n_1} \times \dots \times Z_{n_s}$, provided that $r \geq 0, n_j \geq 2, \forall j$, and $n_{i+1} \mid n_i, \forall 1 \leq i \leq s-1$. (2) G only has one possible (unique) representation of this form. (3) r is called the *free rank* or *Betti number* of G , the integers $n_1, \dots, n_s$ are the *invariant factors* of G , and the representation is called the *invariant factor decomposition* of G .

Fact: (1) Two finitely generated abelian groups are isomorphic $\iff$ same free rank and same list of invariant factors. (2) A finitely generated abelian group is finite only if $r = 0$. (3) Every prime divisor of n must divide the first invariant factor n_1 . (4) The type of G is $(n_1, \dots, n_s)$.

Corollary 4: If n is the product of distinct primes, then up to isomorphism the only abelian group of order n is Z_n .

Fact: Determining all possible isomorphism classes of abelian groups of order n is solved by factoring n and partitioning into invariant factors.

Theorem 5: Let G be an abelian group of order $n > 1$ and let the unique factorization of n into distinct prime powers be $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$. (1) Then $G \cong A_1 \times \dots \times A_k$, where $|A_i| = p_i^{\alpha_i}$ (these are *not* alternating groups). (2) For each $A \in \{A_1, \dots, A_k\}$ with $|A| = p^\alpha$, $A \cong Z_{p^{\beta_1}} \times \dots \times Z_{p^{\beta_t}}$ with $\beta_1 \geq \dots \geq \beta_t \geq 1$ and $\sum_{i=1}^t \beta_i = \alpha$. (3) G has only one possible (unique) decomposition of the form in (1), and each A similarly has only a unique decomposition of the form in (2). Note also that $Z_{p^{\beta_1}} \times \dots \times Z_{p^{\beta_t}}$ are the invariant factors of A_i . (4) The integers p^{β_i} are called the *elementary divisors* of G . The decomposition in (1),(2) is called the *elementary divisor decomposition* of G .

Fact: Determining all the possible invariant factor decompositions of finitely generated abelian groups of order n is equivalent to determining the number of ways to partition each α (for each p^α in the factorization of n), then independently combining them.

Proposition 6: Let $m, n \in \mathbb{Z}^+$. (1) $Z_m \times Z_n \cong Z_{mn} \iff (m, n) = 1$. (2) $n = p_1^{\alpha_1} \dots p_k^{\alpha_k} \implies Z_n \cong Z_{p_1^{\alpha_1}} \times \dots \times Z_{p_k^{\alpha_k}}$.

Example: (Invariant Factors/Cyclic Decomposition $\rightarrow$ Elementary Divisors): $|G| = 1800, G \cong Z_{30} \times Z_{30} \times Z_2 \rightarrow E = \{2, 2, 2, 3, 3, 5, 5\}$.

Example: (Elementary Divisors $\rightarrow$ Invariant Factors): $|G| = 1800, E = \{2, 2, 2, 3, 3, 25\} \rightarrow G \cong Z_{2 \cdot 3 \cdot 25} \times Z_{2 \cdot 3 \cdot 1} \times Z_{2 \cdot 1 \cdot 1} \cong Z_{150} \times Z_6 \times Z_2$.

Example: (Elementary Divisors): (1) $Z_6 \times Z_{15}$ has $E = \{2, 3, 3, 5\} \implies \cong Z_2 \times Z_3 \times Z_3 \times Z_5$. (2) $Z_{10} \times Z_9$ has $E = \{2, 5, 9\} \implies \cong Z_2 \times Z_5 \times Z_9$.

Definition: (1) If G is a finite abelian group of type $(n_1, \dots, n_t)$, then t is called the *rank* of G . (2) The *exponent* of G is the smallest $n \in \mathbb{Z}^+$ such that $x^n = 1, \forall x \in G$. If $\nexists n$, then the exponent is ∞ .

5.4 Recognizing Direct Products

Definition: Let G be a group, $x, y \in G$, and $A, B \subseteq G$ with $A, B \neq \emptyset$. (1) $[x, y] = x^{-1}y^{-1}xy$ is called the *commutator* of x and y . (2) Define $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$. (3) Define $G' = \langle [x, y] \mid x, y \in G \rangle$ to be the *commutator subgroup* of G .

Proposition 7: Let G be a group, $x, y \in G$, and $H \leq G$. (1) $xy = yx[x, y]$. Importantly, $xy = yx \iff [x, y] = 1$. (2) $H \leq G \iff [H, G] < H$. (3) $\sigma[x, y] = [\sigma(x), \sigma(y)]$ for any $\sigma \in \text{Aut}(G)$. Also, G' char G and G/G' is abelian. (4) G/G' is the largest abelian quotient of G , i.e. $H \leq G$ and G/H is abelian $\iff G' \leq H$. (5) If $\varphi : G \rightarrow A$ is any homomorphism of G into an abelian group A , then φ factors through G' , i.e. $G' \leq \ker \varphi$.

Proposition 8: Let $H, K \leq G$. The number of distinct ways of writing each element of the set HK in the form hk is $|H \cap K|$. If $H \cap K = 1$, each element of HK can be written uniquely as hk .

Theorem 9: Let G be a group with $H, K \leq G$ satisfying (1) $H, K \trianglelefteq G$. and (2) $H \cap K = 1$. Then, $HK \cong H \times K$.

Definition: If G is a group satisfying Theorem 9, HK is called the *internal direct product* and $H \times K$ is called the *external direct product*.

5.5 Semidirect Products

Theorem 10: Let H, K be groups and let $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. Let $\cdot$ denote the left action of K on H determined by φ . Let G be the set of (h, k) , then define the following multiplication on G : $(h_1, k_1)(h_2, k_2) = (h_1 \cdot k_1 h_2, k_1 k_2)$. Then, (1) G is a group with order $|H||K|$. (2) $H \cong \{(h, 1) \mid h \in H\}$ and $K \cong \{(1, k) \mid k \in K\}$, and the isomorphic copies of H, K in G are subgroups of G .

Now, identifying H, K with their isomorphic copies in G . (3) $H \trianglelefteq G$. (4) $H \cap K = 1$. (5) $\forall h \in H, \forall k \in K, khk^{-1} = k \cdot h = \varphi(k)(h)$.

Proposition 11: Let H, K be groups and let $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. Then the following are equivalent: (1) $H \rtimes K \cong H \times K$, (2) φ is the trivial homomorphism, (3) $K \trianglelefteq H \rtimes K$.

Theorem 12: Suppose G is a group with $H, K \leq G$ such that (1) $H \trianglelefteq G$ and (2) $H \cap K = 1$. Let $\varphi : K \rightarrow \text{Aut}(H)$ be the homomorphism that maps k to an inner automorphism. Then $HK \cong H \rtimes K$.

Definition: Let $H \leq G$. The *complement* for H in G is some $K \leq G$ with $G = HK$ and $H \cap K = 1$.

Chapter 6: Further Topics in Group Theory

6.1 p -groups, Nilpotent Groups, and Solvable Groups

Definition: A *maximal subgroup* of a group G is a proper subgroup M of G such that $\nexists H \leq G$ with $M < H < G$.

Theorem 1: Let p be prime and P have order p^a , $a > 1$. Then (1) $Z(P) \neq 1$. (2) If H is a nontrivial normal subgroup of P then $H \cap Z(P) \neq 1$. In particular, every normal subgroup of order p is contained in $Z(P)$. (3) If $H \trianglelefteq P$ then $\exists F \leq H$ with $|F| = p^b$ and $F \trianglelefteq P$, $\forall p^b$ that divide $|H|$. In particular, P has a normal subgroup of order p^b for every $b \in \{0, \dots, a\}$. (4) If $H < P$ then $H < N_P(H)$. (5) Every maximal subgroup of P is of index p and is normal in P .

Definition: (1) $Z_0(G) = 1$, $Z_1(G) = Z(G)$, $Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G))$. The chain $Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \dots$ is the *upper central series* of G . (2) G is *nilpotent* if $Z_c(G) = G$ for some $c \in \mathbb{Z}$. The smallest such c is the *nilpotence class* of G .

Proposition 2: Let p be a prime and P be a group of order p^a . Then P is nilpotent of nilpotence class at most $a - 1$ for $a \geq 2$ (and class = a when $a = 0, 1$).

Theorem 3: Let G be a finite group, let $p_1, \dots, p_s$ be the distinct primes dividing $|G|$ and let $P_i \in \text{Syl}_{p_i}(G)$, $1 \leq i \leq s$. Then the following are equivalent: (1) G is nilpotent. (2) If $H < G$ then $H < N_G(H)$. (3) $P_i \trianglelefteq G$ for $1 \leq i \leq s$. (4) $G \cong P_1 \times \dots \times P_s$.

Corollary 4: A finite abelian group is the direct product of its Sylow subgroups.

Proposition 5: If G is a finite group such that for all $n \in \mathbb{Z}^+$ with $n \mid |G|$, G contains at most n elements x satisfying $x^n = 1$, then G is cyclic.

Proposition 6: (*Frattini's Argument*) Let G be a finite group, $H \trianglelefteq G$, and $P \in \text{Syl}_p(H)$. Then $G = HN_G(P)$ and $|G : H|$ divides $|N_G(P)|$.

Proposition 7: A finite group is nilpotent $\iff$ every maximal subgroup is normal.

Definition: For any (finite or infinite) group G we define $G^0 = G$, $G^1 = [G, G]$, $G^{i+1} = [G, G^i]$. The chain of groups $G^0 \geq G^1 \geq G^2 \geq \dots$ is called the *lower central series* of G .

Theorem 8: A group G is nilpotent $\iff G^n = 1$ for some $n \geq 0$. In fact, G is nilpotent of class $c \iff c$ is the smallest nonnegative integer such that $G^c = 1$. Also, then $G^{c-i} \leq Z_i(G)$, $\text{froall } i \in \{0, \dots, c\}$.

Definition: For any group G we define $G^{(0)} = G$, $G^{(1)} = [G, G]$, $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. The chain of groups $G^{(0)} \geq G^{(1)} \geq \dots$ is the *derived or commutator series* of G .

Definition: A group G is *solvable* if there exists a chain of subgroups $1 = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_s = G$ such that G_{i+1}/G_i is abelian

$\forall i \in \{0, 1, \dots, s-1\}$.

Theorem (3.4): A finite group G is solvable $\iff \forall n$ with $n \mid |G|$ and $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n .

Theorem 9: A group G is solvable $\iff G^{(n)} = 1$ for some $n \geq 0$.

Definition: If G is solvable, the smallest nonnegative n such that $G^{(n)} = 1$ is the *solvable length* of G .

Proposition 10: Let G, K be groups, $H \leq G$, and $\varphi : G \rightarrow K$ be a surjective homomorphism. Then (1) $H^{(i)} \leq G^{(i)}$, $\forall i \geq 0$. In particular, if G is solvable, so too is H , and the solvable length of H is $\leq$ the solvable length of G . (2) $\varphi(G^{(i)}) = K^{(i)}$. In particular, homomorphic images and quotient groups of solvable groups are solvable, with solvable length less than or equal to that of the domain group G . (3) If $N \trianglelefteq G$ and both N and G/N are solvable, then so is G .

Theorem 11: Let G be a finite group. (1) (*Burnside*) If $|G| = p^a q^b$ for some primes p, q , then G is solvable. (2) (*Philip Hall*) If $\forall p$ with $p \mid |G|$, we factor $|G| = p^a m$ with $(p, m) = 1$, and G has a subgroup of order m , then G is solvable. (i.e. $\forall p$, G has a subgroup with index equal to the order of a Sylow p -subgroup, called a *Sylow p -complement*, then G is solvable). (3) (*Feit-Thompson*) If $|G|$ is odd then G is solvable. (4) (*Thompson*) If $\forall$ pairs of elements $x, y \in G$, $\langle x, y \rangle$ is solvable, then G is solvable.

6.2 Applications in Groups of Medium Order

Technique 1: (*Counting Elements*) Consider Sylow p -subgroups where $p \mid |G|$ and $p^2 \nmid |G|$. Non-normality implies $n_p > 1$, $\forall p$. Calculate # of elements of prime order from this, show that it is $> |G| \implies$ not simple. Or, show that $n_p = 1$ for some p .

Technique 2: (*Exploiting subgroups of small index*) This is derived from the fact that if $H \leq G$ and $|G : H| = k$, $\exists \varphi : G \rightarrow S_k$ with $\ker \varphi \leq H$. If G is simple, $k = 1 \implies G \cong F \leq S_k \implies |G| \mid k!$ (that is, φ must be an injective homomorphism since $\ker \varphi \trianglelefteq G$ and we assume G is simple).

$|G| = p_1^{\alpha_1} \dots p_s^{\alpha_s}$ with $p_1 < \dots < p_s$. Minimal possible index of a proper subgroup is (usually) dependent on α_s . $\alpha_s = 1 \implies \geq p_s$,

$\alpha_s = 2 \implies \geq 2p_s$.

Technique 3: (*Permutation Representations*) Consider a proper subgroup of index k in G with $|G| = n$. Then, $G \cong F \leq S_k$. We then calculate within S_k that S_k contains no simple subgroup of order n . Two helpful restrictions are (1) if G contains an element/subgroup of a particular order, so too must S_k and (2) if $P \in \text{Syl}_p(G)$ and $P \in \text{Syl}_p(S_k)$, then $|N_G(P)| \mid |N_{S_k}(P)|$. Note that (2) occurs frequently when $k = p$ or $p+1$ and G has a subgroup of index k , as it implies $|N_{S_k}(P)| = p(p-1) \implies |N_G(P)| \mid p(p-1)$. It may also help to work in A_k instead of S_k (see Prop 12).

Technique 4: (*Playing p -subgroups off against each other for distinct primes p*) (1) Suppose p, q are distinct primes such that every group of order pq is cyclic $\implies p \nmid q-1$ with $p < q$. If $Q \in \text{Syl}_q(G)$ with $|Q| = q$ and $p \mid |N_G(Q)|$, $\exists P \leq N_G(Q)$ with $|P| = p$. Thus, PQ is a group, and if PQ is abelian, $PQ \leq N_G(P)$ and thus $q \mid |N_G(P)|$. This may force $N_G(P) = G \implies P \trianglelefteq G$ or at least force $N_G(P)$ to have index less than the minimal index $\implies$ contradiction. (2) We can refine this method. Let $Q \in \text{Syl}_q(G)$ and $p \mid |N_G(Q)|$. Let $P \in \text{Syl}_p(N_G(Q))$. Apply Sylow's Theorem, and if $P \trianglelefteq N_G(Q)$, we force $N_G(P)$ to have small index. If $P \in \text{Syl}_p(G)$, we can put further restrictions on $|N_G(P)|$. If not, then Sylow's Theorem implies $P < P^* \in \text{Syl}_p(G) \implies P < N_{P^*}(P) \leq N_G(P)$.

Technique 5: (*Studying normalizers of intersections of Sylow p -subgroups*) Technique 1 fails to generalize because if distinct $P, R \in \text{Syl}_p(G)$ and $|P| = p^\alpha$, $\alpha \geq 2$, it is not necessary for $P \cap R = 1$. Suppose $\exists P, R$ with the above true. Let $P_0 = P \cap R$. Then, $P_0 < P$ and $P_0 < R$. By Theorem 1, $P_0 < N_P(P_0)$ and $P_0 < N_R(P_0)$. We aim to use this to prove that the normalizer in G has smaller index than the minimal index. This works well when $|P_0| = p^{\alpha-1}$, as then $N_G(P_0)$ has two distinct Sylow p -subgroups P, R since P_0 is a maximal subgroup of P, R . In particular, $|N_G(P_0)| = p^\alpha k$ where $k \geq p + 1$.

Proposition 12: (1) If G has no subgroup of index 2 and $G \leq S_k$, then $G \leq A_k$. (2) If $P \in \text{Syl}_p(S_k)$ for some odd prime p , then $P \in \text{Syl}_p(A_k)$ and $|N_{A_k}(P)| = \frac{1}{2}|N_{S_k}(P)|$.

Lemma 13: In a finite group G , if $n_p \not\equiv 1 \pmod{p^2}$, then $\exists P, R \in \text{Syl}_p(G)$ with $P \neq R$ and $P \cap R$ is of index p in both P, R (and thus normal in each).

Proposition 14: If G is a simple group of order 168, then (1) $n_2 = 21, n_3 = 7, n_7 = 8$. (2) Sylow 2-subgroups of G are dihedral, Sylow 3- and 7-subgroups are cyclic. (3) G is isomorphic to a subgroup of A_7 and G has no subgroup of index ≤ 6 . (4) The conjugacy classes of G are the identity, two classes with order 24 and elements of order 7 (represented by an order 7 element and its inverse), one class with order 56 and elements of order 3, one class of order 42 and elements of order 4, and one class of order 21 and elements of order 2 (in particular, every element of G has order a power of a prime). (5) If $T \in \text{Syl}_2(G)$ and U, W are the two Klein 4-groups in T , then U, W are not conjugate in G and $N_G(U) \cong N_G(W) \cong S_4$. (6) G has precisely three conjugacy classes of maximal subgroups, two $\cong S_4$ and one $\cong$ non-abelian group of order 21.

Theorem 15: Up to isomorphism there is a unique simple group of order 168, $GL_3(\mathbb{F}_2) \cong$ automorphism group of the projective plane $\mathcal{F}$.

6.3 A Word on Free Groups

Theorem 16: $F(S)$ is a group under the binary operation defined as follows:

$$(r_1^{\delta_1} r_2^{\delta_2} \dots r_m^{\delta_m})(s_1^{\delta_1} s_2^{\delta_2} \dots s_m^{\delta_m}) = \begin{cases} r_1^{\delta_1} \dots r_{m-k+1}^{\delta_{m-k+1}} s_k^{\epsilon_k} \dots s_n^{\epsilon_n}, & k \leq m \\ s_{m+1}^{\epsilon_{m+1}} \dots s_n^{\epsilon_n}, & k = m+1 \leq n, \\ 1, & k = m+1, m = n \end{cases}$$

Where k is the smallest integer in the range $1 \leq k \leq m+1$ such that $s_k^{\epsilon_k} \neq r_{m-k+1}^{-\delta_{m-k+1}}$. (Where the product of r and s in the center does not cancel).

Theorem 17: Let G be a group, S a set, and $\varphi : S \rightarrow G$ a set map. Then there is a unique group homomorphism $\Phi : F(S) \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\text{inclusion}} & F(S) \\ & \searrow \varphi & \downarrow \Phi \\ & & G \end{array}$$

Corollary 18: $F(S)$ is unique up to a unique isomorphism which is the identity map on the set S .

Definition: The group $F(S)$ is called the *free group* on the set S . A group F is a *free group* if there is some S such that $F = F(S)$ — in this case we call S a set of *free generators* (or a *free basis*) of F . The cardinality of S is the *rank* of the free group.

Theorem 19: (*Schreier*) Subgroups of a free group are free.

Definition: Let $S \subseteq G$ such that $G = \langle S \rangle$. (1) A *presentation* for G is a pair $\langle S, R \rangle$, where R is a set of words in $F(S)$ such that the normal closure of $\langle R \rangle$ in $F(S)$ (the smallest normal subgroup containing $\langle R \rangle$) equals the $\ker \pi$, where $\pi : F(S) \rightarrow G$ is a homomorphism that extends the identity mapping from S to S . The elements of S are called *generators* and those of R are called *relations*. (2) We say G is *finitely generated* if there is a presentation (S, R) such that S is a finite sets and we say G is *finitely presented* if there is a presentation (S, R) with both S and R finite sets.

Chapter 7: Introduction to Rings

7.1 Basic Definitions and Examples

Definition: (1) A *ring* R is a set together with two binary operations $+$, $\times$, satisfying these axioms: (i) $(R, +)$ is an abelian group. (ii) $\times$ is associative (iii) the *distributive laws* hold in R both ways. (2) The ring R is *commutative* if multiplication is commutative. (3) The ring R is said to have an *identity* if $\exists 1 \in R$ such that $1 \times a = a \times 1 = a, \forall a \in R$.

Definition: A ring R with identity 1 with $1 \neq 0$ is called a *division ring* or *skew field* if $\exists a^{-1}, \forall a \in R$ with $a \neq 0$, such that $aa^{-1} = a^{-1}a = 1$. A *commutative* division ring is called a *field*.

Proposition 1: Let R be a ring and $a, b \in R$. Then (1) $0a = a0 = 0$. (2) $(-a)b = a(-b) = -(ab)$. (3) $(-a)(-b) = ab$. (4) If R has an identity, then the identity is unique and $-a = (-1)a$.

Definition: Let R be a ring. (1) A nonzero element $a \in R$ is called a *zero divisor* if there is a nonzero element $b \in R$ such that *either* $ab = 0$ or $ba = 0$. (2) Assume R has nonzero identity 1 . An element $u \in R$ is called a *unit* if $\exists v \in R$ such that $uv = vu = 1$. The set of units in R is denoted $R^\times$, and is a group under multiplication. Note also that a field is a commutative F with nonzero identity 1 in which every nonzero element is a unit, i.e. $F^\times = F - \{0\}$. Also, a zero divisor can never be a unit.

Definition: A commutative ring with identity $1 \neq 0$ is called an *integral domain* if it has no zero divisors ($\mathbb{Z}$ -like).

Proposition 2: Assume $a, b, c \in R$ with element a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$. In particular, if R is an integral domain, $ab = ac \implies a = 0$ or $b = c$.

Corollary 3: Any finite integral domain is a field. A result that follows is a finite division ring is necessarily commutative, i.e. a field. This won't be proven.

Definition: (*Subring Criterion*) A *subring* of the ring R is a subgroup of R that is closed under multiplication. The subring criterion is showing that it is *nonempty* and *closed under subtraction and under multiplication*.

7.2 Examples: Polynomial Rings, Matrix Rings, and Groups Rings

Polynomial Rings: $R \in R[x]$ as the *constant polynomials*. Addition is done component-wise in $R[x]$. $R[x]$ is a *commutative ring with identity*.

Proposition 4: Let R be an integral domain and let nonzero $p(x), q(x) \in R[x]$. Then (1) $\deg p(x)q(x) = \deg p(x) + \deg q(x)$. (2) the units of $R[x]$ are the units of R . (3) $R[x]$ is an integral domain.

Matrix Rings: $M_n(R)$ is the set of all $n \times n$ matrices with entries from R . If R is nontrivial and $n \geq 2$ then $M_n(R)$ is *non-commutative*. An element $(a_{ij}) \in M_n(R)$ is called a *scalar matrix* if it is diagonal, and the set of scalar matrices is a subring of $M_n(R)$ and is isomorphic to R . If R has an identity 1 , then the identity $I \in M_n(R)$ and the subgroup of units in $M_n(R)$ is $GL_n(R)$. If a ring $S \subseteq R$ then $M_n(S) \subseteq M_n(R)$ (and is also a subring).

Group Rings: Fix a commutative ring R with nonzero identity 1 and let G be a finite group $\{g_1, \dots, g_n\}$. The *group ring* RG of G with coefficients in R to be the set of all formal sums $a_1g_1 + \dots + a_ng_n$, $a_i \in R$. Multiplication is defined by $(ag_i)(bg_j) = (ab)g_k$, where $g_i g_j = g_k$. RG is commutative $\iff G$ is abelian. If $S \subseteq R$ then $SG \subseteq RG$. If $H \leq G$ then $RH \subseteq RG$. The set of all elements of RG whose coefficients sum to zero is a subring without identity. If $|G| > 1$, the set of elements with zero "constant term" (coefficient of identity in G is 0) is *not* a subring (not closed under multiplication).

7.3 Ring Homomorphisms and Quotient Rings

Definition: Let R, S be rings. (1) A *ring homomorphism* is a map $\varphi : R \rightarrow S$ with (i) $\varphi(a+b) = \varphi(a) + \varphi(b)$ and (ii) $\varphi(ab) = \varphi(a)\varphi(b)$. (2) $k \in \ker \varphi \implies \varphi(k) = 0 \in S$. (3) A bijective ring homomorphism is an *isomorphism*.

Proposition 5: Let R, S be rings and $\varphi : R \rightarrow S$ be a homomorphism. (1) $\varphi(R) \leq S$. (2) $\ker \varphi \leq R$. If $k \in \ker \varphi$ then $rk, kr \in \ker \varphi, \forall r \in R$.

Definition: Let R be a ring, $I \subseteq R$, and $r \in R$. (1) $rI = \{ra \mid a \in I\}$ and $Ir = \{ar \mid a \in I\}$. (2) A subset I of R is a *left ideal* of R if $I \leq R$, $rI \subseteq I, \forall r \in R$. A *right ideal* is defined similarly. (3) A subset I that is a left and right ideal is called an *ideal* or *two-sided ideal*.

Fact: (*Ideal Criterion*) I is an ideal if it is nonempty, closed under subtraction, and closed under multiplication by all elements of R .

Proposition 6: Let R be a ring and let I be an ideal of R . Then the additive quotient R/I is a ring under the binary operations $(r+I) + (s+I) = (r+s)+I$ and $(r+I) \times (s+I) = (rs)+I$. The converse holds true too. R/I is a *quotient ring*.

Theorem 7: (1) (*First Isomorphism Theorem*) If $\varphi : R \rightarrow S$ is a homomorphism of rings, $\ker \varphi$ is an ideal of R , $\varphi(R) \leq S$, and $R/\ker \varphi \cong \varphi(R)$. (2) If I is an ideal of R , the map $R \rightarrow R/I$ is a surjective ring homomorphism with kernel I , and is called the *natural projection*. Thus, a subring is an ideal $\iff$ it is the kernel of a ring homomorphism.

Theorem 8: Let R be a ring. (1) (*Second Isomorphism Theorem*) Let $A \leq R$ and B be an ideal of R . Then $A+B = \{a+b \mid a \in A, b \in B\} \leq R$, $A \cap B$ is an ideal of A , and $(A+B)/B \cong A/(A \cap B)$. (2) (*Third Isomorphism Theorem*) Let I, J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and $(R/I)/(J/I) \cong R/J$. (3) (*Fourth/Lattice Isomorphism Theorem*) Let I be an ideal of R . The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings A of R that contain I and the set of subrings of R/I . And, A is an ideal of $R \iff A/I$ is an ideal of R/I .

Definition: Let I, J be ideals of R . $I+J = \{a+b \mid a \in I, b \in J\}$, $IJ = \{\sum ab \mid a \in I, b \in J\}$. $I^n = \{\sum a_1 \dots a_n \mid a_i \in I, 1 \leq i \leq n\}$.

7.4 Properties of Ideals

Definition: Let $A \subseteq R$. (1) Let (A) denote the smallest ideal of R containing A , called the *ideal generated by A* . (2) Let $RA = \{r_1a_1 + \dots + r_na_n \mid r_i \in R, a_i \in A\}$, and AR and RAR be defined similarly. ($RA = 0$ if $A = \emptyset$). (3) An ideal generated by a single element is a *principal ideal* (like a cyclic group). (4) An ideal generated by a finite set is a *finitely generated ideal*. (5) If R is commutative then $RA = AR = RAR = (A)$.

Proposition 9: Let I be an ideal of R . (1) $I = R \iff I$ contains a unit. (2) Assume R is commutative. R is a field $\iff$ its only ideals are 0 and R .

Corollary 10: If R is a field then any nonzero ring homomorphism from R into another ring is an injection.

Definition: An ideal M in a ring S is *maximal* if $M \neq S$ and the only ideals containing M are M and S .

Proposition 11: In a ring with identity every proper ideal is contained in a maximal ideal.

Proposition 12: Assume R is commutative. The ideal M is maximal $\iff R/M$ is a field.

Definition: Assume R is commutative. An ideal P is *prime* if $P \neq R$ and whenever $ab \in P$ when $a, b \in R$, at least one of $a, b \in P$.

Proposition 13: Assume R is commutative. Then the ideal P is a prime ideal in $R \iff R/P$ is an integral domain.

Corollary 14: Assume R is commutative. Every maximal ideal of R is prime

7.5 Rings of Fractions

Theorem 15: Let R be a commutative ring. Let D be any nonempty subset of R that does not contain 0, does not contain any zero divisors, and is closed under multiplication (i.e., $ab \in D, \forall a, b \in D$). Then there is a commutative ring Q with 1 such that $R \leq Q$ and $\forall d \in D, d$ is a unit in Q . The ring Q has the following additional properties. (1) Every element of Q is of the form rd^{-1} for some $r \in R$ and $d \in D$. In particular, if $D = R - \{0\}$, then Q is a field. (2) (uniqueness of Q) The ring Q is the *smallest* ring containing R in which all elements of D become units: Let S be a commutative ring with identity and $\varphi : R \rightarrow S$ be an injective ring homomorphism such that $\varphi(d)$ is a unit in $S, \forall d \in D$. Then $\exists \Phi : Q \rightarrow S$ that is an injective homomorphism such that $\Phi|_R = \varphi$. That is, any ring containing some $R' \cong R$ in which all elements of D become units must also contain some $Q' \cong Q$.

Definition: Let R, D, Q be as in Theorem 15. (1) Q is denoted the *ring of fractions* of D with respect to R and is denoted $D^{-1}R$. (2) If R is an integral domain and $D = R - \{0\}$, Q is the *field of fractions* or *quotient field* of R .

Fact: If $A \subseteq F$, the intersection of all subfields of F containing A is a subfield of F and is denoted the subfield *generated by A* . This subfield is the smallest subfield of F containing A , where size is defined analogously as in Theorem 15.

Corollary 16: Let R be an integral domain and let Q be the field of fractions of R . If a field F contains a subring $R' \cong R$ then the subfield of F generated by R' is isomorphic to Q .

7.6 The Chinese Remainder Theorem

Definition: The ideals A, B of the ring R are *comaximal* if $A + B = R$.

Theorem 17: (Chinese Remainder Theorem) Let $A_1, \dots, A_k$ be ideals in R . The map $R \rightarrow R/A_1 \times \dots \times R/A_k$ defined by $r \mapsto (r + A_1, \dots, r + A_k)$ is a ring homomorphism with kernel $\bigcap_{i=1}^k A_i$. If $\forall i, j \in \{1, \dots, k\}$ with $i \neq j$ we have $A_i + A_j = R$, then the map is surjective and $\bigcap_{i=1}^k A_i = A_1 \dots A_k = \prod_{i=1}^k A_i$. Thus, $R/(A_1 \dots A_k) = R/(A_1 \cap \dots \cap A_k) \cong R/A_1 \times \dots \times R/A_k$.

Corollary 18: Let $n \in \mathbb{Z}^+$ and let $p_1^{a_1} \dots p_k^{a_k}$ be its prime factorization. Then $\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{a_1}\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_k^{a_k}\mathbb{Z})$ as rings. In particular, for multiplicative groups, $(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{a_1}\mathbb{Z})^\times \times \dots \times (\mathbb{Z}/p_k^{a_k}\mathbb{Z})^\times$.

Chapter 8: Euclidean Domains, Principal Ideal Domains, and Unique Factorization Domains

8.1 Euclidean Domains

Definition: Any function $N : R \rightarrow \mathbb{Z}^+ \cup \{0\}$ with $N(0) = 0$ is called a *norm* on the integral domain R . If $N(a) > 0$, $\forall a \neq 0$, then N is called a *positive norm*.

Definition: The integral domain R is said to be a *Euclidean Domain* or possess a *Division Algorithm* if $\exists N$ a norm on R such that $\forall a, b \in R$ with $b \neq 0$, $\exists q, r \in R$ such that $a = qb + r$ and either $r = 0$ or $N(r) < N(b)$. q is the *quotient* and r is the *remainder*.

Proposition 1: Every ideal I in a Euclidean Domain R is principal. More precisely, if $I \neq \emptyset$ then $I = (d)$, where $d \in I$, $d \neq 0$, and $N(d)$ is minimal.

Definition: Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. (1) a is a *multiple* of b if $\exists x \in R$ such that $a = bx$. Then b *divides* a and is a *divisor* of a . (2) A *greatest common divisor* of a, b is some $d \neq 0$ such that (i) $d \mid a$ and $d \mid b$ and (2) if $d' \mid a$ and $d' \mid b$ then $d' \mid d$.

Fact: If I is an ideal in R generated by a, b , then $d = (a, b)$ if (i) $I \subseteq (d)$ and (ii) if $I \subseteq (d')$ then $(d) \subseteq (d')$.

Proposition 2: If $a, b \in R$ such that $a, b \neq 0$ and $(a, b) = (d)$, then $d = (a, b)$.

Proposition 3: Let R be an integral domain. Let $d, d' \in R$ such that $(d) = (d')$. Then, $\exists u \in R$ that is a unit such that $d' = ud$. In particular, if d, d' are both GCDs of a, b , then $d' = ud$ for some unit u . (GCD is not necessarily unique).

Fact: For any $a, b \in R$ a Euclidean Domain, (a, b) always exists and can be computed algorithmically (Euclidean Algorithm).

Theorem 4: Let R be a Euclidean Domain and $a, b \in R$ such that $a, b \neq 0$. Let $d = r_n$ be the last nonzero remainder in the Euclidean Algorithm for a, b . Then (1) $d \in (a, b)$, the set of GCDs, and (2) $(d) = (a, b)$, i.e. d can be written as an R -linear combination of a, b , or $\exists x, y \in R$ such that $d = ax + by$. (Bezout's Identity).

Fact: The equation $ax + by = N$ is solvable in integers $x, y \iff (a, b) \mid N$.

Fact: For any integral domain let $\tilde{R} = R^\times \cup \{0\}$ denote the collection of units of R together with 0. An element $u \in R - \tilde{R}$ is called a *universal side divisor* if $\forall x \in R$ then $\exists z \in \tilde{R}$ such that $u \mid x - z$ in R , i.e. there is a type of "division algorithm" for u such that $x = qu + z$ for some $q \in R$. The existence of universal side divisors is a weakening of the Euclidean condition.

Proposition 5: Let R be an integral domain that is not a field. If R is a Euclidean Domain then there are universal side divisors in R .

8.2 Principal Ideal Domains (P.I.D.s)

Definition: A *Principal Ideal Domain* is an integral domain in which every ideal is principal.

Proposition 6: Let R be a PID and let $a, b \in R$ such that $a, b \neq 0$. Let $(d) = (a, b)$. Then (1) $d \in (a, b)$. (2) $\exists x, y \in R$ such that $d = ax + by$. (3) d is unique up to multiplication by $u \in R^\times$ (unit).

Proposition 7: Every nonzero prime ideal in a PID is a maximal ideal.

Corollary 8: If R is any commutative ring such that the polynomial ring $R[x]$ is a PID then R is a field.

Definition: N is a *Dedekind-Hasse norm* if N is a positive norm and $\forall a, b \in R$ such that $a, b \neq 0$ either $a \in (b)$ or $\exists c \in (a, b)$ such that $N(c) < N(b)$. In other words, either $b \mid a$ or $\exists s, t \in R$ with $0 < N(sa - tb) < N(b)$.

Proposition 9: The integral domain R is a PID $\iff R$ has a Dedekind-Hasse norm.

8.3 Unique Factorization Domains (U.F.D.s)

Definition: Let R be an integral domain. (1) Suppose $r \in R$ is nonzero and not a unit. Then r is *irreducible* in R if whenever $r = ab$ with $a, b \in R$, at least one of a, b must be a unit in R . Otherwise, r is *reducible*. (2) The nonzero element $p \in R$ is *prime* in R if (p) is a prime ideal. That is, $p \neq 0$ is prime if it is not a unit and, whenever $p \mid ab$ for any $a, b \in R$, then either $p \mid a$ or $p \mid b$. (3) Any $a, b \in R$ such that $a = ub$ for some unit $u \in R$, then a, b are considered *associate*.

Proposition 10: In an integral domain a prime element is always irreducible. The converse is, in general, not true.

Proposition 11: In a PID a nonzero element is a prime $\iff$ it's irreducible.

Definition: A *Unique Factorization Domain* is an integral domain R in which $\forall r \in R$ with $r \neq 0$ which is not a unit satisfies: (1) r can be written as a finite product of not necessarily distinct irreducibles $p_i \in R$, i.e. $r = p_1 \dots p_n$. (2) the decomposition is *unique up to associates*, namely, if $r = q_1 \dots q_m$ is another valid factorization into irreducibles, then $m = n$ and there is a bijective map with $p_i \mapsto q_j$ such that p_i is associate with q_j . **Proposition 12:** In a UFD a nonzero element is a prime $\iff$ it is irreducible. (Stronger version of proposition 11).

Proposition 13: Let $a, b \in R$ a UFD with $a, b \neq 0$ and suppose $a = up_1^{e_1} \dots p_n^{e_n}$ and $b = vp_1^{f_1} \dots p_n^{f_n}$ are prime factorizations where u, v are units, the primes are distinct, and $e_i, f_i \geq 0$. Then $d = p_1^{\min(e_1, f_1)} \dots p_n^{\min(e_n, f_n)}$ is the gcd.

Theorem 14: $\{\text{Fields}\} \subset \{\text{Euclidean Domains}\} \subset \{\text{PID}\} \subset \{\text{UFD}\} \subset \{\text{Integral Domains}\}$.

Corollary 15: (*Fundamental Theorem of Arithmetic*) $\mathbb{Z}$ is a UFD.

Corollary 16: Let R be a PID. Then there exists a multiplicative Dedekind-Hasse norm on R .

Lemma 17: The prime number $p \in \mathbb{Z}$ divides some $n^2 + 1 \in \mathbb{Z} \iff p = 2$ or $p \equiv 1 \pmod{4}$.

Proposition 18: (1) (*Fermat's Theorem on sums of squares*) Let p be a prime. $p = a^2 + b^2$ for $a, b \in \mathbb{Z} \iff p = 2$ or $p \equiv 1 \pmod{4}$. The representation is unique up to changing signs or interchanging a, b . (2) The irreducible elements in the Gaussian integers $\mathbb{Z}[i]$ as follows: (a) $1 + i$

(norm = 2) (b) primes $p \in \mathbb{Z}$ with $p \equiv 3 \pmod{4}$ (norm = p^2). (3) $a + bi, a - bi$, i.e. the distinct irreducible factors of $p = a^2 + b^2$, for primes $p \in \mathbb{Z}$ with $p \equiv 1 \pmod{4}$ (norm = p).

Corollary 19: Let n be a positive integer. Then, we can write $n = 2^k p_1^{a_1} \dots p_r^{a_r} q_1^{b_1} \dots q_s^{b_s}$ where $p_1, \dots, p_r \equiv 1 \pmod{4}$ and are distinct primes and $q_1, \dots, q_s \equiv 3 \pmod{4}$ and are distinct primes. Then n can be written as a sum of two squares in $\mathbb{Z} \iff$ each b_i is even. Then, the number of distinct representations of n as a sum of two squares is $4(a_1 + 1) \dots (a_r + 1)$

Chapter 9: Polynomial Rings

9.1 Definitions and Basic Properties

Proposition 1: Let R be an integral domain. Then (1) $\deg p(x)q(x) = \deg p(x) + \deg q(x)$ if $p(x), q(x) \neq 0$. (2) the units of $R[x]$ are just the units of R . (3) $R[x]$ is an integral domain.

Proposition 2: Let I be an ideal of the ring R and let $(I) = I[x]$ denote the ideal of $R[x]$ generated by I (polynomials with coefficients in I). Then $R[x]/(I) \cong (R/I)[x]$. In particular, if I is a prime ideal of R then (I) is a prime ideal of $R[x]$.

Fact: If I is as described above and I is a maximal ideal of R , $(I, x) \subseteq R[x]$ is a maximal ideal too. (I) is not necessarily maximal in $R[x]$.

Definition: The polynomial ring in variables $x_1, \dots, x_n$ with coefficients in R is denoted $R[x_1, \dots, x_n]$ and defined inductively by

$$R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n].$$

Fact: The degree of a multivariate monomial is the largest sum of the degrees of each variable taken individually. The degree of a multivariate polynomial is the maximum degree of its monomials. The subsum of a polynomial f including only the monomial terms of degree k is denoted the *homogeneous component of f of degree k* . If f has degree d , it is possible to express $f = \sum_{i=0}^d f_i$ for homogeneous components f_i of degree i .

9.2 Polynomial Rings over Fields I

Theorem 3: Let F be a field. The polynomial ring $F[x]$ is a Euclidean Domain. That is, letting $a(x), b(x) \in F[x]$ with $b(x) \neq 0$, then there exist unique $q(x), r(x) \in F[x]$ such that $a(x) = q(x)b(x) + r(x)$ with $r(x) = 0$ or $\deg r(x) < \deg b(x)$. ($\deg$ is the norm).

Corollary 4: If F is a field, then $F[x]$ is a PID and UFD (since it is already a Euclidean Domain).

9.3 Polynomial Rings that are Unique Factorization Domains

Proposition 5: (Gauss's Lemma) Let R be a UFD with field of fractions F and let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$ then $p(x)$ is reducible in $R[x]$. More precisely, if $p(x) = A(x)B(x)$ for some nonconstant polynomials $A(x), B(x) \in F[x]$, then $\exists r, s \in F$ with $r, s \neq 0$ such that $rA(x) = a(x)$ and $sB(x) = b(x)$, $a(x), b(x) \in R[x]$, and $p(x) = a(x)b(x)$ is a factorization in R .

Corollary 6: Let R be a UFD, F be its field of fractions, and $p(x) \in R[x]$. Suppose the gcd of the coefficients of $p(x)$ is 1. Then $p(x)$ is irreducible in $R[x] \iff$ it is irreducible in $F[x]$. In particular, if $p(x)$ is monic and irreducible in $R[x]$ then $p(x)$ is irreducible in $F[x]$. An important implication of this is that a monic polynomial in $\mathbb{Z}[x]$ is irreducible $\iff$ it is irreducible over $\mathbb{Q}[x]$. (Can also be applied to polynomials in $\mathbb{Q}[x]$ with only integer coefficients).

Theorem 7: R is a UFD $\iff R[x]$ is a UFD.

Corollary 8: If R is a UFD, then a multivariate polynomial ring with coefficients in R is also a UFD.

9.4 Irreducibility Criteria

Proposition 9: Let F be a field and let $p(x) \in F[x]$. Then $p(x)$ has a factor of degree 1 $\iff \exists \alpha \in F$ with $p(\alpha) = 0$.

Proposition 10: A polynomial with $\deg = 2$ or 3 over a field F is reducible $\iff$ it has a root in F .

Proposition 11: (Rational Root Theorem) Let $p(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x]$. If $\frac{r}{s} \in \mathbb{Q}$ in the lowest terms and $p(\frac{r}{s}) = 0$, then r divides the constant term and s divides the leading coefficient of $p(x)$: $r \mid a_0$ and $s \mid a_n$. In particular, if $p(x)$ is *monic* with integer coefficients and $p(d) \neq 0, \forall d \in \mathbb{Z}$ such that $d \mid a_0$, then $p(x)$ has no roots in $\mathbb{Q}$. More precisely, if $p(x) \in \mathbb{Z}[x]$ and is monic, all rational roots must be integers.

Proposition 12: Let I be a proper ideal in the integral domain R and let $p(x)$ be a nonconstant monic polynomial in $R[x]$. If the image of $p(x)$ in $(R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of smaller degree, then $p(x)$ is irreducible in $R[x]$.

Proposition 13: (Eisenstein's Criterion) Let P be a prime ideal of the integral domain R and let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in R[x]$ ($n \geq 1$). Suppose $a_{n-1}, \dots, a_0 \in P$ and $a_0 \notin P^2$. Then $f(x)$ is irreducible in $R[x]$.

Corollary 14: (Eisenstein's Criterion for $\mathbb{Z}[x]$) Let prime $p \in \mathbb{Z}$ and $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$, $n \geq 1$. Suppose $p \mid a_i, \forall i \in \{0, \dots, n-1\}$ but $p^2 \nmid a_0$. Then $f(x)$ is irreducible in both $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

9.5 Polynomial Rings over Fields II

Let F be a field.

Proposition 15: The maximal ideals in $F[x]$ are the ideals $(f(x))$ generated by irreducible polynomials $f(x)$. In particular, $F[x]/(f(x))$ is a field $\iff f(x)$ is irreducible.

Proposition 16: Let $g(x)$ be a nonconstant monic element of $F[x]$ and let $g(x) = f_1(x)^{n_1} \dots f_k(x)^{n_k}$ be its irreducible factorization. Then we have the following ring isomorphism: $F[x]/(g(x)) \cong F[x]/(f_1(x)^{n_1}) \times \dots \times F[x]/(f_k(x)^{n_k})$.

Proposition 17: If the polynomial $f(x)$ has roots $\alpha_1, \dots, \alpha_k \in F$ (not necessarily distinct), then $(x - \alpha_1) \dots (x - \alpha_k) \mid f(x)$. In particular, a univariate polynomial of degree n over F has at most n roots in F , even with multiplicity.

Proposition 18: A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Corollary 19: Let p be a prime. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero residue classes mod p is cyclic. (Since $(\mathbb{Z}/p\mathbb{Z})^\times$ is a finite field).

Corollary 20: Let $n \geq 2$ be an integer with factorization $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ in $\mathbb{Z}$, where $p_1, \dots, p_r$ are distinct primes. Then, the following isomorphisms of multiplicative groups exist: (1) $(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})^\times$. (2) $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times \cong Z_2 \times Z_{2^{\alpha-2}}, \forall \alpha \geq 2$. (3) $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times \cong Z_{p^{\alpha-1}(p-1)}$, for all odd primes p .

9.6 Polynomials in Several Variables over a Field and Gröbner Bases

Definition: A commutative ring R with 1 is called *Noetherian* if every ideal of R is finitely generated.

Theorem 21: (*Hilbert's Basis Theorem*) If R is a *Noetherian* ring then so is the polynomial ring $R[x]$.

Corollary 22: Every ideal in the polynomial ring $F[x_1, \dots, x_n]$ with coefficients from a field F is finitely generated.

Definition: A *monomial ordering* is a well ordering " $\geq$ " on the set of monomials that satisfies $mm_1 \geq mm_2$ whenever $m_1 \geq m_2$ for monomials m, m_1, m_2 . Equivalently, a monomial ordering may be specified by defining a well ordering on the n -tuples $\alpha = (a_1, \dots, a_n) \in \mathbb{Z}^n$ of multidegrees of monomials $Ax_1^{\alpha_1} \cdots x_n^{\alpha_n}$ that satisfies $\alpha + \gamma \geq \beta + \gamma$ if $\alpha \geq \beta$.

Definition: Fix a monomial ordering on the polynomial ring $F[x_1, \dots, x_n]$. (1) The *leading term* of a nonzero polynomial $f \in F[x_1, \dots, x_n]$, denoted $LT(f)$, is the monomial term of maximal order in f and the leading term of $f = 0$ is 0. Define the *multidegree* of f , denoted $\partial(f)$, to be the multidegree of the leading term of f . (2) If I is an ideal in $F[x_1, \dots, x_n]$, the *ideal of leading terms*, denoted $LT(I)$, is the ideal generated by the leading terms of all the elements in the ideal, i.e., $LT(I) = (LT(f) \mid f \in I)$.

Fact: $LT(fg) = LT(f) + LT(g)$. The ideal $LT(I)$ is also generated by monomials; such ideals are denoted as *monomial ideals*. A polynomial is contained in a monomial ideal $\iff$ each of its monomial terms is a multiple of one of the ideal's generators.

Definition: A *Gröbner basis* for an ideal I in the polynomial ring $F[x_1, \dots, x_n]$ is a finite set of generators $\{g_1, \dots, g_m\}$ for I whose leading terms generate the ideal of all leading terms in I , i.e. $I = (g_1, \dots, g_m)$ and $LT(I) = (LT(g_1), \dots, LT(g_m))$.

Fact: Every element in I is a linear combination of the generators that is *not necessarily unique*.

General Polynomial Division: Fix a monomial ordering on $F[x_1, \dots, x_n]$ and suppose $g_1, \dots, g_m$ is a set of nonzero polynomials in $F[x_1, \dots, x_n]$. If $f \in F[x_1, \dots, x_n]$, start with a set of quotients $q_1, \dots, q_m$ and a remainder r all equal to 0. Then, recursively test if the leading term of f is divisible by the any of the leading terms of the divisors $g_1, \dots, g_m$ in that order. (i) If $LT(g_i) \mid LT(f)$ such that $LT(f) = a_i LT(g_i)$, add a_i to quotient q_i and repeat the algorithm on $f - a_i g_i$. (ii) If the leading term of the dividend f is not divisible by any of the leading terms $LT(g_1), \dots, LT(g_m)$, add the leading term of f to the remainder r and repeat the algorithm on $f - LT(f)$.

Theorem 23: Fix a monomial ordering on $R = F[x_1, \dots, x_n]$ and suppose $\{g_1, \dots, g_m\}$ is a Gröbner basis for the nonzero ideal I in R . Then (1) Every polynomial $f \in R$ can be written uniquely in the form $f_I + r$ where $f_I \in I$ and no nonzero monomial term of the remainder r is divisible by any of the leading terms $LT(g_1), \dots, LT(g_m)$. (2) Both f_I and r can be computed by general polynomial division by $g_1, \dots, g_m$ and are independent of the order in which these polynomials are used in the division. (3) The remainder r provides a unique representative for the coset of f in $F[x_1, \dots, x_n]/I$. In particular, $f \in I \iff r = 0$.

Proposition 24: Fix a monomial ordering on $R = F[x_1, \dots, x_n]$ and let I be a nonzero ideal in R . (1) If $g_1, \dots, g_m \in I$ such that $LT(I) = (LT(g_1), \dots, LT(g_m))$. (2) There exists a Gröbner basis for I .

Fact: Let M be the monic LCM of $LT(f_1)$ and $LT(f_2)$. Define $S(f_1, f_2) = \frac{M}{LT(f_1)} f_1 - \frac{M}{LT(f_2)} f_2$. Then $S(f_1, f_2)$ is a polynomial where the leading terms of f_1 and f_2 have been canceled.

Lemma 25: Suppose $f_1, \dots, f_m \in F[x_1, \dots, x_n]$ are polynomials with the same multidegree α and that the linear combination $h = a_1 f_1 + \cdots + a_m f_m$ with constants $a_i \in F$ has a multidegree strictly smaller than α . Then $h = \sum_{i=2}^m b_i S(f_{i-1}, f_i)$ for some constants $b_i \in F$.

Fact: For a fixed monomial ordering on $R = F[x_1, \dots, x_n]$ and ordered set of polynomials $G = \{g_1, \dots, g_m\}$ in R , write $f \equiv r \pmod{G}$ if r is the remainder of general polynomial division of $f \in R$ by $g_1, \dots, g_m$ in that order.

Proposition 26: (*Buchberger's Criterion*) Let $R = F[x_1, \dots, x_n]$ and fix a monomial ordering on R . If $I = \{g_1, \dots, g_m\}$ is a nonzero ideal in R , then $G = \{g_1, \dots, g_m\}$ is a Gröbner basis for $I \iff S(g_i, g_j) \equiv 0 \pmod{G}, 1 \leq i < j \leq m$.

Buchberger's Algorithm: Let ideal $I = (g_1, \dots, g_m)$ and $G = \{g_1, \dots, g_m\}$. Let $S(g_i, g_j)$ have $r \neq 0$ for some ordered pair (i, j) . (Otherwise, if $S(g_i, g_j) = 0, \forall (i, j)$, then G is a Gröbner basis). Modify G to be $G' = G \cup \{r\}$. Iterate until done. Note that this will always terminate after a finite number of sets since there exist finitely many ordered pairs (i, j) .

Fact: If $G = \{g_1, \dots, g_m\}$ is a Gröbner basis for the ideal I and $LT(g_i) \mid LT(g_j)$ for some $j \neq i$, then $G - \{g_j\}$ is still a valid Gröbner basis. Therefore, we may assume that the $LT(g_i), \forall i$ is monic. Then, a Gröbner basis $G = \{g_1, \dots, g_m\}$ for I is called *minimal* if $LT(g_i)$ is monic $\forall i$ and $LT(g_i) \nmid LT(g_j), \forall (i, j)$. (This is analogous to a linear independent basis in vector spaces). Note also that while a minimal Gröbner basis is not necessarily unique, the number of elements and their leading terms are unique.

Definition: Fix a monomial ordering on $R = F[x_1, \dots, x_n]$. A Gröbner basis $\{g_1, \dots, g_m\}$ for a nonzero ideal I in R is called a *reduced Gröbner basis* if (a) $LT(g_i)$ is monic $\forall i$ and (b) no term in g_j is divisible by $LT(g_i), \forall (i, j)$ where $i \neq j$.

Fact: A reduced Gröbner basis is also clearly minimal.

Theorem 27: Fix a monomial ordering on $R = F[x_1, \dots, x_n]$. Then there is a *unique* reduced Gröbner basis for every nonzero ideal I in R .

Corollary 28: Let I, J be ideals in $F[x_1, \dots, x_n]$. Then $I = J \iff I$ and J have the same reduced Gröbner basis with respect to any fixed monomial ordering on $F[x_1, \dots, x_n]$.

Definition: If I is an ideal in $F[x_1, \dots, x_n]$ then $I_i = I \cap F[x_{i+1}, \dots, x_n]$ = set of polynomials in I that do not involve the variables $x_1, \dots, x_i$ is called the *i th elimination ideal* of I with respect to the lexicographic monomial ordering $x_1 > \cdots > x_n$.

Proposition 29: (*Elimination*) Suppose $G = \{g_1, \dots, g_m\}$ is a Gröbner basis for the nonzero ideal I in $F[x_1, \dots, x_n]$ with respect to the above ordering. Then $G \cap F[x_{i+1}, \dots, x_n]$ is a Gröbner basis of the *i th elimination ideal* defined above. In particular,

$$I \cap F[x_{i+1}, \dots, x_n] = 0 \iff G \cap F[x_{i+1}, \dots, x_n] = \emptyset.$$

Fact: Let I, J be ideals in $F[x_1, \dots, x_n]$ with $I = (f_1, \dots, f_s)$ and $J = (h_1, \dots, h_t)$. Then $I + J = (f_1, \dots, f_s, h_1, \dots, h_t)$ and $IJ = (f_1 h_1, \dots, f_i h_j, \dots, f_s h_t)$.

Proposition 30: IF I, J are ideals in $F[x_1, \dots, x_n]$ then $tI + (1 - t)J$ is an ideal in $F[t, x_1, \dots, x_n]$ and $I \cap J = (tI + (1 - t)J) \cap F[x_1, \dots, x_n]$. In particular, $I \cap J$ is the first elimination ideal of $tI + (1 - t)J$ with respect to the ordering $t > x_1 > \dots > x_n$.

Chapter 13: Field Theory

13.1 Basic Theory of Field Extensions

Definition: The *characteristic* of a field F , denoted $\text{ch}(F)$, is the smallest $p \in \mathbb{Z}^+$ such that $p \cdot 1_F = 0$ if such a p exists. Otherwise, $\text{ch}(F) = 0$.

Proposition 1: $\text{ch}(F)$ is either 0 or a prime p . If $\text{ch}(F) = p$ then for any $\alpha \in F$, $\alpha p = 0$.

Fact: This notion of a characteristic makes sense for an integral domain too, whose characteristic will be the same as for its fields of fractions.

Definition: The *prime subfield* of a field F is the subfield of F generated by the multiplicative identity 1_F of F . It is isomorphic to $\mathbb{Q}$ if $\text{ch}(F) = 0$ or $\mathbb{F}_p$ if $\text{ch}(F) = p$.

Definition: If K is a field containing the subfield F , then K is said to be an *extension field* or simply an *extension* of F , denoted K/F or by the diagram

$$\begin{array}{c} K \\ | \\ F \end{array}$$

In particular, every field is an extension of its prime subfield. The field F is sometimes called the *base field* of the extension.

Fact: If K/F is any extension of fields, then the multiplication defined in K makes K a *vector space* over F . In particular every field F can be considered a vector space over its prime field.

Definition: The *degree* (or *relative degree* or *index*) of a field extension K/F , $[K : F]$, is the dimension of K as a vector space over F (i.e., $[K : F] = \dim_F K$). The extension is said to be *finite* if $[K : F]$ is finite and is otherwise *infinite*.

Proposition 2: Let $\varphi : F \rightarrow F'$ be a field homomorphism. Then φ is either identically 0 or is injective, i.e. $\varphi(F) \cong \{0\}$ or $\varphi(F) \cong F$.

Theorem 3: Let F be a field and let $p(x) \in F[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of F in which $p(x)$ has a root. Identifying F with this isomorphic copy shows that there exists an extension of F in which $p(x)$ has a root.

Theorem 4: Let $p(x) \in F[x]$ be an irreducible polynomial of degree n over the field F and let K be the field $F[x]/(p(x))$. Let $\theta = x \pmod{(p(x))} \in K$. Then the elements $1, \theta, \dots, \theta^{n-1}$ are a basis for K as a vector space over F , so $[K : F] = n$. Hence

$K = \{a_0 + a_1\theta + \dots + a_{n-1}\theta^{n-1} \mid a_0, a_1, \dots, a_{n-1} \in F\}$, i.e. all polynomials of degree $< n$ in θ .

Corollary 5: Let K be as in Theorem 4 and let $a(\theta), b(\theta) \in K$ be two polynomials of degree $< n$ in θ . Then addition in K is analogous to usual polynomial addition and multiplication in K is defined by $a(\theta)b(\theta) = r(x)$, where $r(x) = a(x)b(x) \pmod{p(x)}$.

Definition: Let K be an extension of the field F and let $\alpha, \beta, \dots \in K$. Then the smallest subfield of K containing both F and the elements $\alpha, \beta, \dots$ is called the field *generated by* $\alpha, \beta, \dots$ over F and denoted $F(\alpha, \beta, \dots)$.

Definition: If the field K is generated by a single element α over F , $K = F(\alpha)$, then K is said to be a *simple extension* of F and the element α is called a *primitive element* for the extension.

Fact: Every finite extension of a field of characteristic 0 is a simple extension.

Theorem 6: Let F be a field and let $p(x) \in F[x]$ be irreducible. Suppose K extends F and contains a root α of $p(x)$. Then $F(\alpha) \cong F[x]/(p(x))$. In particular, *any* field over F in which $p(x)$ contains a root must contain a subfield isomorphic to the extension of F constructed in Theorem 3 and that this field is, up to isomorphism, the smallest extension of F containing such a root.

Corollary 7: Suppose in Theorem 6 that $\deg p(x) = n$. Then $F(\alpha) = \{a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1} \mid a_0, a_1, \dots, a_{n-1} \in F\} \subseteq K$.

Fact: The roots of an irreducible polynomial $p(x) \in F[x]$ are *algebraically indistinguishable* in the sense that the fields obtained by adjoining any root of an irreducible polynomial are isomorphic.

Theorem 8: Let $\varphi : F \xrightarrow{\sim} F'$ be a field isomorphism. Let $p(x) \in F[x]$ be irreducible and let $p'(x) \in F'[x]$ be the polynomial obtained by applying φ to the coefficients of $p(x)$. Let α be a root of $p(x)$ in some extension of F and let β be a root of $p'(x)$ in some extension of F' . Then there exists an isomorphism $\sigma : F(\alpha) \xrightarrow{\sim} F'(\beta)$ such that $\alpha \mapsto \beta$ and such that σ extends φ , i.e. σ restricted to F is the isomorphism φ . Pictorially, it is

$$\begin{array}{ccccc} \sigma : & F(\alpha) & \xrightarrow{\sim} & F'(\beta) & \\ & | & & | & \\ \varphi : & F & \xrightarrow{\sim} & F' & \end{array}$$

13.2 Algebraic Extensions

Let F be a field and K be an extension of F .

Definition: The element $\alpha \in K$ is *algebraic* over F if $f(\alpha) = 0$ for some nonzero $f(x) \in F[x]$. If α is not algebraic over F , then α is *transcendental* over F . K/F is *algebraic* if every element of K is algebraic over F .

Fact: If α is algebraic over F then it is trivially algebraic over any extension field L of F .

Proposition 9: Let α be algebraic over F . Then there exists a unique monic irreducible polynomial $m_{\alpha, F}(x) \in F[x]$ for which α is a root.

$f(x) \in F[x]$ has $f(\alpha) = 0 \iff m_{\alpha, F}(x) \mid f(x)$ in $F[x]$.

Corollary 10: If L/F is an extension of fields and α is algebraic over both F and L , then $m_{\alpha, L}(x) \mid m_{\alpha, F}(x)$ in $L[x]$.

Definition: The polynomial $m_{\alpha, F}(x)$ is called the *minimal polynomial* for α over F . The *degree* of $m_{\alpha}(x)$ is also called the *degree* of α .

Proposition 11: Let α be algebraic over F . Then $F(\alpha) \cong F[x]/(m_{\alpha}(x))$, and thus $[F(\alpha) : F] = \deg m_{\alpha}(x) = \deg \alpha$, i.e. the degree of α over F is the degree of the extension it generates over F .

Proposition 12: The element α is algebraic over $F \iff$ the simple extension $F(\alpha)/F$ is finite. More precisely, if α is an element of an extension

of degree n over F then α is a root of a polynomial of degree at most n over F , and if α satisfies a polynomial of degree n over F then the degree of $F(\alpha)$ over F is at most n .

Corollary 13: If the extension K/F is finite, then it is algebraic.

Fact: Any extension K of F such that $[K : F] = 2$ is of the form $F(\sqrt{D})$, where $D \in F$ such that D is not a square in F , and conversely every extension of the form $F(\sqrt{D})$ has degree 2. These are known as the *quadratic extensions* of F .

Theorem 14: Let $F \subseteq K \subseteq L$ be fields. Then $[L : F] = [L : K][K : F]$, i.e. extension degrees are multiplicative, and if one side of the equation is infinite, the other side is also infinite. Pictorially,

$$\overbrace{F \subseteq K \subseteq L}^{[L:F]} \quad \overbrace{\quad \quad \quad}^{[L:F]} \quad \underbrace{F \subseteq K}_{[K:F]} \subseteq \underbrace{K \subseteq L}_{[L:K]}$$

Corollary 15: Suppose L/F is a finite extension and let $F \subseteq K \subseteq L$. Then $[K : F] \mid [L : F]$.

Definition: An extension K/F is *finitely generated* if $\exists \alpha_1, \dots, \alpha_k \in K$ such that $K = F(\alpha_1, \dots, \alpha_k)$.

Lemma 16: $F(\alpha, \beta) = (F(\alpha))(\beta)$, i.e. the field generated over F by α and β is the field generated by β over the field $F(\alpha)$ generated by α .

Fact: The field $F(\alpha)$ for algebraic α is typically obtained by adjoining α to F and then closing the resulting set with respect to addition and multiplication. The process terminates when a power of α is a linear combination of lower powers of α , i.e. finding the minimal polynomial for α .

If α is non-algebraic, it is necessary to also close with respect to additive and multiplicative inverse.

Theorem 17: The extension K/F is finite $\iff$ K is finitely generated by algebraic elements over F . More precisely, let $\alpha_1, \dots, \alpha_k \in F$ be algebraic with degrees $n_1, \dots, n_k$. Then $\deg F(\alpha_1, \dots, \alpha_k) \leq n_1 \dots n_k$.

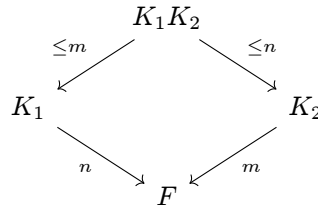
Corollary 18: Suppose α, β are algebraic over F . Then $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ (for $\beta \neq 0$) are all algebraic.

Corollary 19: Let L/F be an arbitrary extension. Then the set of elements of L that are algebraic over F form a subfield K of L .

Theorem 20: If K is algebraic over F and L is algebraic over K , then L is algebraic over F .

Definition: Let $K_1, K_2 \subseteq K$. Then the *composite field* of K_1 and K_2 , denoted $K_1 K_2$, is the smallest subfield of K with $K_1, K_2 \subseteq K_1 K_2$. This extends to any number of subfields.

Proposition 21: Let K_1, K_2 be two finite extensions of a field $F \subseteq K$. Then $[K_1 K_2 : F] \leq [K_1 : F][K_2 : F]$. Equality is true $\iff$ an F -basis for one of the fields remains linearly independent over the other field. If $\{\alpha_1, \dots, \alpha_n\}$ and $\{\beta_1, \dots, \beta_m\}$ are the bases for K_1 and K_2 , respectively, then the elements $\alpha_i \beta_j$ span $K_1 K_2$ over F . Pictorially,



Corollary 22: Suppose that $[K_1 : F] = n, [K_2 : F] = m$ in Proposition 21, where $(m, n) = 1$. Then $K_1 K_2 : F = [K_1 : F][K_2 : F] = mn$.

13.3 Classical Straightedge and Compass Constructions

Proposition 23: If the element $\alpha \in \mathbb{R}$ is obtained from a field $F \subset \mathbb{R}$ by a series of compass and straightedge constructions then $[F(\alpha) : F] = 2^k$ for some integer $k \geq 0$.

Theorem 24: None of the classical Greek problems: (I) Doubling the Cube, (II) Trisecting an Angle, and (III) Squaring the Circle, is possible.

13.4 Splitting Fields and Algebraic Closures

Let F be a field.

Definition: The extension field K of F is called a *splitting field* for the polynomial $f(x) \in F[x]$ if $f(x)$ factors (*splits*) completely into linear factors in $K[x]$ and $f(x)$ does not factor completely over any proper subfield of K containing F .

Theorem 25: If $f(x) \in F[x]$ then there exists an extension K of F which is a splitting field for $f(x)$.

Fact: If $f(x) \in F[x]$ such that $\deg f(x) = 1$, then the splitting field of F for $f(x)$ is simply F . If $\deg f(x) = 2$ and $f(x)$ is irreducible, then the algebraic extension $F(\alpha)$ over F for some root α of $f(x)$ contains all roots of $f(x)$. If $\deg f(x) \geq 3$, then the splitting field may be a chain of algebraic extensions.

Definition: If K is an algebraic extension of F which is the splitting field over F for a collection of polynomials $f(x) \in F[x]$ then K is called a *normal* extension of F .

Proposition 26: A splitting field of a polynomial of degree n over F is of degree at most $n!$ over F .

Definition: A generator of the cyclic group of all n th roots of unity is called a *primitive* n th root of unity ζ_n .

Definition: The field $\mathbb{Q}(\zeta_n)$ is called the *cyclotomic field of n th roots of unity*.

Fact: The polynomial $\frac{x^p - 1}{x - 1}$ is the minimal polynomial for ζ_p and has degree $\varphi(p) = p - 1$. In general, $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$.

Theorem 27: Let $\varphi : F \rightarrow F'$ be a field isomorphism. Let $f(x) \in F[x]$ be a polynomial and let $f'(x) \in F'[x]$ be the polynomial obtained by applying

φ to the coefficients of $f(x)$. Let E be a splitting field for $f(x)$ over F and let E' be a splitting field for $f'(x)$ over F' . Then the isomorphism φ extends to an isomorphism $\sigma : E \xrightarrow{\sim} E'$, i.e. σ restricted to F is φ . Pictorially,

$$\begin{array}{ccc} \sigma : & E & \xrightarrow{\sim} E' \\ & \downarrow & \downarrow \\ \varphi : & F & \xrightarrow{\sim} F' \end{array}$$

Corollary 28: (Uniqueness of Splitting Fields) Any two splitting fields for a polynomial $f(x) \in F[x]$ over a field F are isomorphic.

Definition: The field $\overline{F}$ is called an *algebraic closure* of F if $\overline{F}$ is algebraic over F and if every polynomial $f(x) \in F[x]$ splits completely over $\overline{F}$ (that is, $\overline{F}$ contains all the elements algebraic over F).

Definition: A field K is *algebraically closed* if every polynomial with coefficients in K has a root in K .

Proposition 29: Let $\overline{F}$ be an algebraic closure of F . Then $\overline{F}$ is algebraically closed.

Proposition 30: For any field F there exists an algebraically closed field K containing F .

Proposition 31: Let K be an algebraically closed field and let F be a subfield of K . Then the collection of elements $\overline{F}$ of K that are algebraic over F is an algebraic closure of F . An algebraic closure of F is unique up to isomorphism.

Theorem: (Fundamental Theorem of Algebra) The field $\mathbb{C}$ is algebraically closed.

Corollary 32: The field $\mathbb{C}$ contains an algebraic closure for any of its subfields. In particular, $\overline{\mathbb{Q}}$, the collection of complex numbers algebraic over $\mathbb{Q}$, is an algebraic closure of $\mathbb{Q}$.